

Advanced Intelligent Ticket Fraud Detection and Risk-Aware Passenger Verification

R26-IT-127

Project Proposal Report

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B.Sc. (Hons) Degree in Information Technology Specialized in
Information Technology

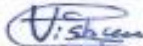
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March 2026

DECLARATION

I declare that this is my own work, and this proposal does not knowingly incorporate any material previously submitted for a degree or diploma at any other university or higher education institution, nor does it contain any material that has been previously published or written by another person apart from where proper acknowledgment is given in the text.

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ABSTRACT

Railway transportation in Sri Lanka is a major mode of transportation for millions of passengers daily. It is also one of the most cost-effective and environmentally friendly modes of transportation for the public. Nevertheless, the present method of ticket inspection in Sri Lanka Railways is based on manual inspection by train inspectors. This method involves a random inspection of passenger tickets. It is quite challenging for the railway authorities to identify complex ticket fraud patterns like ticket reuse, unusual concession usage, or ticket resale activities.

Although data analytics-based or machine learning-based fraud detection systems have been proposed for different sectors like transportation, their application for railway ticketing systems or railway environments is quite rare. Moreover, fraud detection problems often have imbalanced data sets where fraud cases occur less often. In this case, supervised learning methods cannot be used effectively since there are fewer fraud instances.

To address these challenges, this research proposes the Advanced Intelligent Ticket Fraud Detection & Risk Aware Passenger Verification framework that employs unsupervised & semi-supervised anomaly detection methods for identifying unusual ticketing patterns. The proposed framework for detecting ticket fraud in railway systems involves the development of a comprehensive railway fraud taxonomy that includes ticket reuse, unusual concession usage, and unusual patterns of resale. Based on this taxonomy, a set of synthetic or anonymized ticket transaction data sets that mimic realistic passenger behavior patterns will be created for the purpose of model development & testing.

The proposed framework will employ anomaly detection methods like Isolation Forests, One-Class Support Vector Machines, Autoencoders, etc., for identifying unusual patterns in ticket usage. A risk calculation component will be integrated into the proposed framework for measuring the severity of anomalies & identifying key features that contributed to the anomaly, thus enabling railway inspectors to identify high-risk passengers during ticket inspection.

To measure the efficiency of the proposed framework, precision, recall, F1-score, and ROC-AUC will be used, along with a measure of the trade-off between fraud detection efficiency and inspector workload. The proposed system will help in enhancing the efficiency of fraud detection by changing the current inspection style from a random approach to a more risk-based approach for more efficient railway ticket inspection.

Keywords: Railway Fraud Detection, Anomaly Detection, Risk Scoring, Ticketing Analytics, Explainable AI

ACKNOWLEDGEMENT

I would like to express my heartfelt gratitude to my research supervisor for the guidance and valuable suggestions provided to me during this research. The supervisor's guidance and suggestions helped shape this research and improve its quality. I would also like to thank the academic staff of the Sri Lanka Institute of Information Technology (SLIIT) for providing the necessary knowledge and academic environment to complete this research successfully. My heartfelt appreciation goes to my friends and colleagues for their guidance and encouragement received during the development of this research component. Lastly, I would like to express my deepest gratitude to my family for their encouragement, patience, and support received during my academic pursuit.

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LIST OF ABBREVIATIONS

Abbreviation	Full Form
AI	Artificial Intelligence
ML	Machine Learning
SVM	Support Vector Machine
OCSVM	One-Class Support Vector Machine
ROC-AUC	Receiver Operating Characteristic – Area Under the Curve
F1-score	Harmonic Mean of Precision and Recall
IoT	Internet of Things
MQTT	Message Queuing Telemetry Transport
GPS	Global Positioning System
SDG	Sustainable Development Goal
WBS	Work Breakdown Structure
UI	User Interface
IDE	Integrated Development Environment
CPU	Central Processing Unit
RAM	Random Access Memory
SSD	Solid State Drive
HDD	Hard Disk Drive
AWS	Amazon Web Services
API	Application Programming Interface
ML Model	Machine Learning Model

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1. INTRODUCTION

1.1 Background and Context

Railway transportation is one of the most widely used modes of public transportation in Sri Lanka, providing an affordable and accessible means of travel for a large portion of the population. Thousands of passengers rely on railway services daily for commuting between cities and suburban areas. Due to its affordability and convenience, railway transportation plays a significant role in supporting daily mobility and economic activities. However, despite its importance, railway systems face several operational challenges, particularly in ticket management and fraud prevention.

One of the major challenges in railway transportation systems is ticket fraud and fare evasion. Fraudulent activities such as ticket reuse, unauthorized concession usage, and ticket resale can lead to revenue losses and operational inefficiencies. In many railway systems, ticket verification processes still rely on manual inspection conducted by train inspectors. This manual approach makes it difficult to consistently detect complex fraudulent behaviors and identify suspicious passengers efficiently. Previous studies on railway ticketing systems have highlighted the limitations of traditional ticket verification processes and the need for more secure and automated solutions [1], [2].

The impact of ticket fraud extends beyond financial losses. Railway operators may experience reduced revenue generation and inefficiencies in resource allocation, while inspectors face difficulties in identifying suspicious passengers due to the lack of analytical tools to support decision-making. In addition, honest passengers may experience inconvenience because inspections are often conducted randomly without considering the likelihood of fraud. Research on behavior-based fraud detection has emphasized the importance of monitoring user activities and identifying abnormal patterns to improve operational efficiency and fraud identification [3].

Globally, fare evasion and ticket fraud are recognized as significant issues in public transportation systems. Studies indicate that fare evasion can account for approximately 3% to 10% of transportation revenue losses in some public transport networks. As passenger volumes increase, detecting fraudulent ticket usage becomes more challenging, particularly in systems where inspection mechanisms remain largely manual. With the increasing digitalization of transportation services, researchers have proposed various technological solutions to improve ticket security and verification processes. For example, blockchain-based ticketing systems have been proposed to enhance ticket security and prevent tampering through decentralized transaction management [4].

Several studies have explored intelligent passenger verification systems aimed at improving ticket inspection processes. For instance, artificial intelligence-based verification methods have been proposed to automate passenger identification and improve verification accuracy [5]. Similarly, facial recognition-based ticket verification systems have been developed to digitize the ticket inspection process and reduce manual errors [6]. Other systems integrate ticket verification with seat allocation and passenger authentication mechanisms to streamline railway management operations [7], [8]. While these approaches improve passenger verification and ticket authentication, they primarily focus on identity validation rather than detecting abnormal ticket usage patterns or fraud behaviors..

Further, several studies have focused on data-driven safety and monitoring in railway environments. For example, machine learning has been used for passenger detection to reduce railway accidents [9], while big data analytics has been applied to improve passenger safety in railway stations [10]. Although these studies show the importance of intelligent technologies in railway systems, they do not directly address ticket fraud detection based on passenger behavioral patterns.

To address these limitations, this research proposes an Advanced Intelligent Ticket Fraud Detection and Risk-Aware Passenger Verification framework. The proposed approach begins with the development of a comprehensive railway fraud taxonomy to categorize common fraudulent behaviors such as ticket reuse, abnormal concession usage, and suspicious ticket resale activities. Based on this taxonomy, synthetic or anonymized ticket transaction datasets will be generated to simulate realistic passenger travel behaviors. Behavioral, temporal, and contextual features will then be extracted from ticket transactions to represent passenger travel patterns. These features will be analyzed using anomaly detection algorithms including Isolation Forest, One-Class Support Vector Machine (One-Class SVM), and Autoencoder-based models to identify suspicious ticketing behaviors. In addition, a risk scoring and explanation mechanism will be developed to quantify the severity of detected anomalies and highlight the key factors contributing to suspicious behaviors, enabling inspectors to prioritize high-risk passengers during ticket verification.

The proposed research aligns with several United Nations Sustainable Development Goals (SDGs). It contributes to SDG 9 – Industry, Innovation and Infrastructure by introducing intelligent, data-driven technologies to enhance the management and efficiency of railway transportation systems. Furthermore, it supports SDG 11 – Sustainable Cities and Communities by improving the fairness, transparency, and efficiency of public transportation services. From a research perspective, the proposed study aligns with the SST – Software Systems and Technologies research cluster. This research area focuses on the development of intelligent software systems that utilize advanced computing techniques such as machine learning, data analytics, and decision support systems. By integrating anomaly detection algorithms, data engineering methods, and explainable risk-scoring mechanisms, the proposed system aims to develop a scalable software solution for detecting ticket fraud and supporting risk-aware passenger verification in railway transportation systems

.1.2 Literature Review

The rapid development of railway transport systems has led to an increased need for effective ticket management, passenger verification, and fraud detection systems. The traditional process of ticket inspection mainly depends on manual verification, which is not effective in identifying sophisticated ticket misuse patterns. Recent studies have proposed various technology-based solutions for improving railway ticketing and verification processes, including blockchain-based ticketing systems, artificial intelligence-based passenger verification systems, smart ticket inspection systems, and data-driven safety management systems.

Some recent studies have focused on improving the security of railway ticketing systems. Farhan et al. proposed a blockchain-based railway ticketing framework that utilizes QR codes and smart contract technology to increase transparency and security in ticketing processes [1]. The proposed system stores ticket transaction information in a decentralized ledger, thereby reducing the risk of fraudulent modification of ticket records. In another study, Singh and Singh presented a fraud detection approach based on monitoring customer behavior and activity patterns to identify fraudulent actions [2]. Their work demonstrates that deviations in normal behavioral patterns can be useful for identifying suspicious activities, although such methods are generally applied in non-transport domains and often depend on known fraud behavior.

Passenger verification has also been widely explored as a way to improve railway ticket inspection. Kirme et al. proposed an AI-based passenger verification system that uses intelligent mechanisms for passenger identity verification during journeys [3]. Patil et al. developed a digitized railway ticket verification system using facial recognition to improve verification accuracy and reduce manual errors [4]. Gobhinath et al. proposed the Digirail prototype, which combines ticket verification with seat allocation functions for railway passengers [5]. Similarly, Mhamane and Shriram introduced a railway ticket verification and dynamic seat allocation method using Aadhar card integration [6]. These studies demonstrate the growing use of digital technologies in passenger authentication and railway service management. However, their main focus is on identity verification and operational convenience rather than identifying abnormal ticket usage behavior or fraud patterns.

In addition to verification systems, some studies have proposed improved smart ticket inspection and management solutions. Nawghare et al. introduced a smart ticketing validation system for Indian Railways with an improved alert mechanism to help identify illegal ticketing activities during inspection [7]. Likewise, Reddy et al. proposed a foolproof ticketing system management approach aimed at reducing ticket misuse through improved ticket management and inspection procedures [8]. Although these systems strengthen ticket validation processes, they mainly focus on secure ticket issuance and inspection rather than on detecting fraud through behavioral, temporal, and contextual analysis of passenger ticketing data.

Beyond ticketing and verification, several studies have explored intelligent monitoring and safety management in railway environments. Gorane and Joshi proposed a machine learning-based passenger detection system to reduce railway accidents by detecting passengers in railway areas [9]. Alawad et al. examined the use of big data analytics to improve passenger safety in railway stations [10]. These studies highlight the value of intelligent technologies in railway systems, but their primary focus is on safety monitoring and risk reduction rather than fraud detection in ticketing transactions.

From the existing literature, it is evident that several advancements have been made in secure ticketing systems, automated passenger verification systems, smart ticket validation, and data-driven railway safety management . However, these studies are mainly focused on ticket issuance security, passenger identity verification, safety improvement, and operational management. Limited attention has been given to detecting ticket fraud based on passenger behavioral patterns. Existing systems do not adequately address fraud types such as ticket reuse, abnormal concession usage, and suspicious resale behavior through anomaly-based analysis. Therefore, there is a clear need for an intelligent ticket fraud detection system that can analyze ticket transaction patterns, identify abnormalities, and support risk-aware passenger verification in railway transportation systems.

1.3 Research Gap

In spite of these developments in ITS and machine learning-based analytics, several critical gaps exist in the domain of railway ticket fraud detection.

First, most transportation analytics systems are designed with a focus on enhancing efficiency in transportation operations, such as tracking, scheduling, and information systems, among others. While these systems are useful for enhancing efficiency in railway operations, they are not designed for handling issues related to detecting fraudulent activities in passenger transactions.

Second, most existing approaches for detecting fraudulent activities in transportation analytics are based on machine learning algorithms, particularly those that are supervised in nature, where a dataset containing both fraudulent and legitimate transactions is necessary for training a machine learning model for detection purposes. In most cases, fraudulent activities in railways are few, and a dataset for training a machine learning model may not be available, making it difficult for a machine learning-based approach to be implemented in transportation analytics systems.

Third, existing railway ticket inspection processes are based on manual and random processes of checking passengers' tickets, whereby inspectors do not have any analytical tools to determine which passengers are more prone to fraudulent activities. Such processes are not only inefficient but also fail to detect complex patterns of fraudulent activities.

The second limitation is the fact that existing systems are not based on incorporating behavioral, temporal, or contextual features derived from the ticketing process. For example, the travel patterns of passengers or their concession usage can be an important source of information regarding suspicious activities. However, existing systems are not based on incorporating such behavioral indicators for detecting fraudulent activities.

Moreover, existing research is not based on incorporating explainable mechanisms for detecting fraudulent activities. Explainability is an important feature for any system to be adopted for practical

purposes since inspectors need to understand the reason for flagging a particular passenger as suspicious before taking any action against them.

Lastly, there is a lack of studies on the trade-off between the efficiency of fraud detection and the workload of the inspectors. While increasing the intensity of inspections may enhance the rate of detecting fraud, it may also have a negative impact on the overall cost of operations. Additionally, it may reduce the efficiency of the inspection process. Most existing studies have concentrated on the efficiency of the detection without considering its impact on the overall inspection process.

To address the aforementioned limitations of the existing body of knowledge on the problem under study, this research aims to introduce a proactive and explainable anomaly detection system for the efficient detection of railway ticket fraud. The proposed system will incorporate a combination of unsupervised and semi-supervised anomaly detection algorithms such as Isolation Forests, One-Class Support Vector Machines, Autoencoders, among others. Additionally, the proposed system will incorporate a risk scoring mechanism to emphasize the level of anomalies detected as well as the features contributing to the anomalies.

The specific novelty of the current study is the assessment of the trade-off between the performance of fraud detection and the workload of inspectors. The aforementioned topic offers a useful and operational perspective that is not commonly discussed in transportation fraud detection studies. The novelty of the proposed framework is that it introduces a new data-driven approach for intelligent railway ticket fraud detection and verification.

2. OBJECTIVE

2.1 Main Objective

To develop an intelligent railway ticket fraud detection and risk-aware passenger verification system that identifies suspicious ticketing behaviors using anomaly detection techniques and supports railway inspectors through a risk-scoring decision support mechanism.

2.2 Specific Objectives

1. Development of a Railway Fraud Taxonomy and Dataset Preparation

To design a comprehensive railway ticket fraud taxonomy that categorizes fraudulent behaviors such as ticket reuse, abnormal concession usage, and suspicious ticket resale, and to generate synthetic or anonymized railway ticket transaction datasets that simulate realistic passenger travel patterns for model development and evaluation.

2. Feature Engineering for Passenger Behavior Analysis

To extract and construct behavioral, temporal, and contextual features from railway ticket transaction data, including travel frequency patterns, time-based travel characteristics, concession usage patterns, and route-based travel behaviors, in order to represent normal passenger travel patterns for anomaly detection.

3. Implementation of Anomaly Detection Models for Fraud Identification

To implement unsupervised and semi-supervised anomaly detection models including Isolation Forest, One-Class Support Vector Machine (One-Class SVM), and Autoencoder-based models to identify abnormal ticket transaction patterns that may indicate fraudulent ticket usage.

4. Development of a Risk-Scoring and Explanation Framework

To develop a risk-scoring framework that assigns risk levels to ticket transactions based on anomaly detection outputs and provide explanation mechanisms that highlight the key features contributing to suspicious behaviors to support inspector decision-making during passenger verification.

5. Evaluation of Fraud Detection Performance and Inspector Workload Trade-Off

To evaluate the effectiveness of the proposed fraud detection framework using performance metrics such as precision, recall, F1-score, and ROC-AUC, and to analyze the trade-off between fraud detection accuracy and inspector workload by examining different risk-threshold scenarios.

3. METHODOLOGY

3.1 Research Approach

This research adopts a Design Science Research (DSR) approach to develop an intelligent software-based framework for railway ticket fraud detection and risk-aware passenger verification. Design Science Research focuses on designing and evaluating technological artifacts that solve real-world problems. In this study, the artifact is a data-driven fraud detection system that analyzes railway ticket transaction data to identify suspicious passenger behaviors.

The research process involves several stages including fraud taxonomy development, dataset preparation, feature engineering, anomaly detection model implementation, and risk-scoring mechanism development. The proposed system will then be evaluated using established machine learning evaluation metrics to assess its effectiveness in identifying fraudulent ticket transactions while supporting efficient inspection processes.

3.2 Technical Methods and Algorithms

The proposed framework utilizes unsupervised and semi-supervised anomaly detection techniques to identify abnormal ticket transaction patterns. Fraud detection in transportation systems is challenging because fraudulent cases are relatively rare and labeled fraud datasets are often unavailable. Therefore, techniques capable of identifying anomalies based on deviations from normal behavioral patterns are required.

Previous research has shown that monitoring user behavior and activity patterns can effectively identify fraudulent activities in transactional systems [2]. Similarly, intelligent ticket verification and automated passenger authentication systems have been introduced in railway environments to improve ticket validation and operational efficiency [3], [4]. Smart ticket validation mechanisms with automated alerts have also been proposed to improve railway ticket inspection processes [7]. These studies highlight the importance of intelligent data-driven techniques in railway systems but mainly focus on ticket validation and identity verification rather than detecting abnormal behavioral patterns in ticket usage.

Furthermore, research in railway analytics demonstrates the value of large-scale data analysis in improving monitoring and operational decision-making within railway systems [10].

Building upon these insights, the proposed study applies anomaly detection algorithms to analyze ticket transaction behavior and identify suspicious patterns that may indicate fraudulent activities.

Isolation Forest

Isolation Forest is an anomaly detection algorithm that identifies unusual observations by isolating them using randomly generated decision trees. Observations that require fewer splits to be isolated are considered anomalies. This algorithm is efficient for detecting rare abnormal patterns in large datasets and is widely used in fraud detection tasks where anomalous events are sparse.

One-Class Support Vector Machine (One-Class SVM)

One-Class Support Vector Machine is a semi-supervised learning method used when training data primarily consists of normal observations. The algorithm learns a boundary that encloses normal data points and identifies observations outside the boundary as anomalies. This method is particularly useful in fraud detection scenarios where labeled fraudulent data is limited.

Autoencoder-Based Anomaly Detection

Autoencoders are neural network models used for unsupervised representation learning. The model learns to reconstruct normal input data by encoding it into a lower-dimensional representation and reconstructing it back to the original form. When abnormal data is processed by the model, reconstruction errors increase significantly. These reconstruction errors are used to identify suspicious or anomalous transactions.

By combining these anomaly detection techniques, the proposed system can effectively analyze passenger ticket transaction patterns and detect abnormal behaviors that may indicate ticket fraud.

3.4 Data Collection and Ethical Considerations

Due to privacy restrictions associated with real passenger data, this study will utilize synthetic or anonymized railway ticket transaction datasets that simulate realistic passenger travel behavior. The dataset will be generated based on the railway fraud taxonomy defined in this research and will include both normal travel patterns and simulated fraudulent activities such as ticket reuse, abnormal concession usage, and suspicious ticket resale.

Using anonymized or synthetic datasets ensures that passenger privacy and ethical data handling principles are maintained while enabling the development and testing of fraud detection models.

3.5 Validation Strategy and Performance Metrics

The effectiveness of the proposed fraud detection framework will be evaluated using standard machine learning evaluation metrics. These include:

- **Precision** – measures the proportion of detected fraud cases that are actually fraudulent
- **Recall** – measures the ability of the model to correctly identify fraudulent transactions
- **F1-Score** – provides a balanced measure of precision and recall
- **ROC-AUC** – evaluates the model's ability to distinguish between normal and fraudulent transactions

In addition to these metrics, the study will analyze the trade-off between fraud detection performance and inspector workload. Different risk score thresholds will be tested to determine how many passengers are flagged for inspection under different scenarios. This evaluation will help identify an optimal balance between fraud detection accuracy and practical inspection efficiency.

4. HIGH-LEVEL SYSTEM ARCHITECTURE

4.1 System Overview Diagram

The proposed system, named AI-Driven Intelligent Railway Operations System, is intended to assist railway operators, passengers, and transportation authorities by employing artificial intelligence, machine learning, and data analytics technologies to enhance the efficiency of railway operations and the overall experience for passengers. The railway transportation sector in Sri Lanka is also facing some challenges in accurately estimating the demand for passengers, unexpected delays in trains, seat allocation issues, ticket fraud, and static scheduling for trains. The proposed system will address these challenges by developing an intelligent system that analyzes the data for railway operations and offers predictive insights for better management in the sector.

The overall system comprises four intelligent components that monitor and optimize different aspects of railway operations.

The Passenger Demand Forecasting and Adaptive Seat Allocation Component uses historical ticketing information, seasonality, and travel trends to forecast the demand for passengers on different routes within a railway system. Using machine learning algorithms, the system predicts the number of passengers expected on different routes within a given time frame. This prediction is then used by the system to allocate seats for passengers, thereby optimizing seat allocation and utilization.

The Real-Time Train Tracking and Delay Prediction Component uses GPS, timestamps, and timetable information to track trains and estimate their delays. This component uses machine learning algorithms and sequence prediction models to estimate the real-time delays of trains, thus providing accurate information on train delays and estimated times of arrival for passengers and operators within a railway system.

The Ticket Fraud Detection and Risk Analysis Component tracks ticketing transactions and passenger pattern behaviors to identify suspicious activity such as ticket reuse, abnormal concession usage, and unauthorized ticket resale. By applying anomaly detection algorithms and machine learning models, the system identifies abnormal ticketing behaviors. This enables ticket inspectors to identify areas to focus on to minimize revenue loss due to fraudulent activity.

The Train Scheduling Optimization and Capacity Management Component uses passenger demand forecasts, delay patterns, and operational constraints to generate optimized train schedules. By applying multi-objective optimization algorithms, the system evaluates alternative train schedules to identify efficient schedules. This enables railway planners to make informed train scheduling decisions.

In general terms, the AI-Driven Intelligent Railway Operations System is a digital platform that combines all these intelligent systems in a way that helps railway authorities monitor passenger demand, predict delays, prevent ticket fraud, and optimize scheduling. The idea behind this system is that by using data from transportation systems along with artificial intelligence technology, railway reliability can be improved, passenger satisfaction increased, and operational inefficiencies reduced

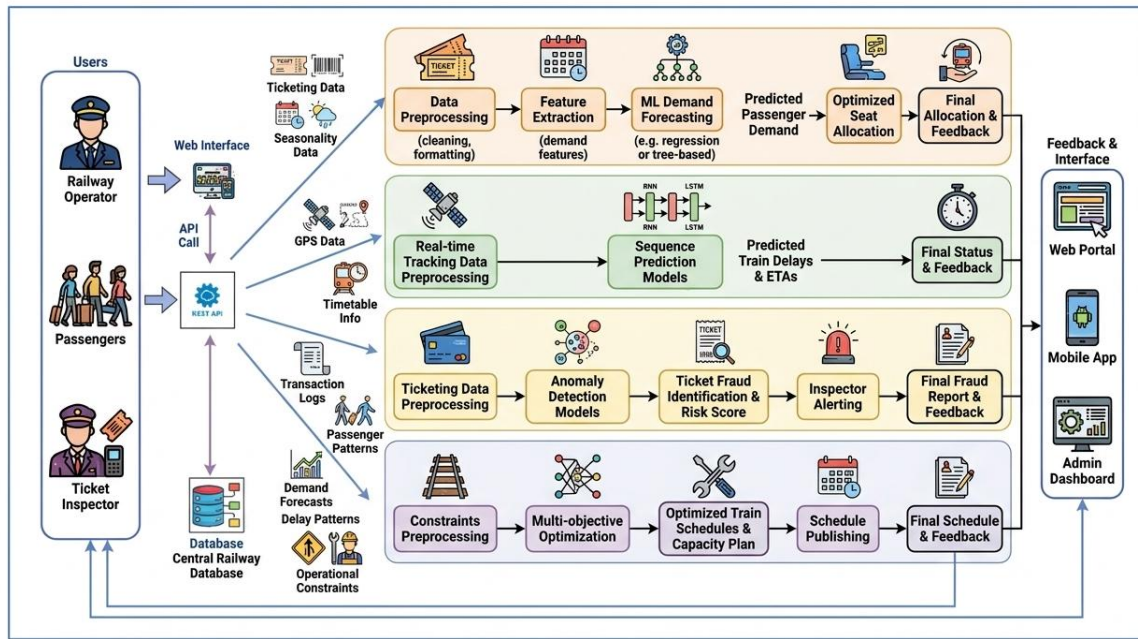


Figure 1: System overview diagram

4.2 Component Overview

The proposed system provides a framework for Advanced Intelligent Ticket Fraud Detection and Risk-Aware Passenger Verification that has the ability to detect unusual patterns in ticketing activities for railway transportation systems. The proposed system utilizes anomaly detection methods for passenger ticket transaction data analysis, aiming to assist railway inspectors in decision-making.

The proposed system comprises several components that interact with each other for data collection, analysis, and generation of results for inspection purposes. The proposed system starts by collecting ticket transaction data from railway ticketing systems. This data may include details on the time taken for a ticket purchase, route details, patterns of usage, and any other relevant information from the railway ticketing systems. After data collection, the data is then transformed into a suitable format for analysis.

Subsequently, feature engineering is carried out, where relevant behavioral, temporal, and contextual features are derived from ticket transactions. These features are a representation of normal passenger

travel behavioral patterns, which are then fed into machine learning algorithms for anomaly detection purposes. Finally, machine learning algorithms are applied to analyze the preprocessed data, where Isolation Forest, One-Class SVM, and Autoencoder algorithms are used for detecting abnormal transactional patterns.

Once abnormal patterns are detected, a risk-scoring framework is employed for assessing the severity of abnormal transactions in the system. In this case, a risk score is assigned to each ticket transaction, as well as features that are responsible for the abnormality in each transaction. The end results of the system include passenger transactions with a risk score, as well as features that aid in identifying suspicious passengers during ticket verification

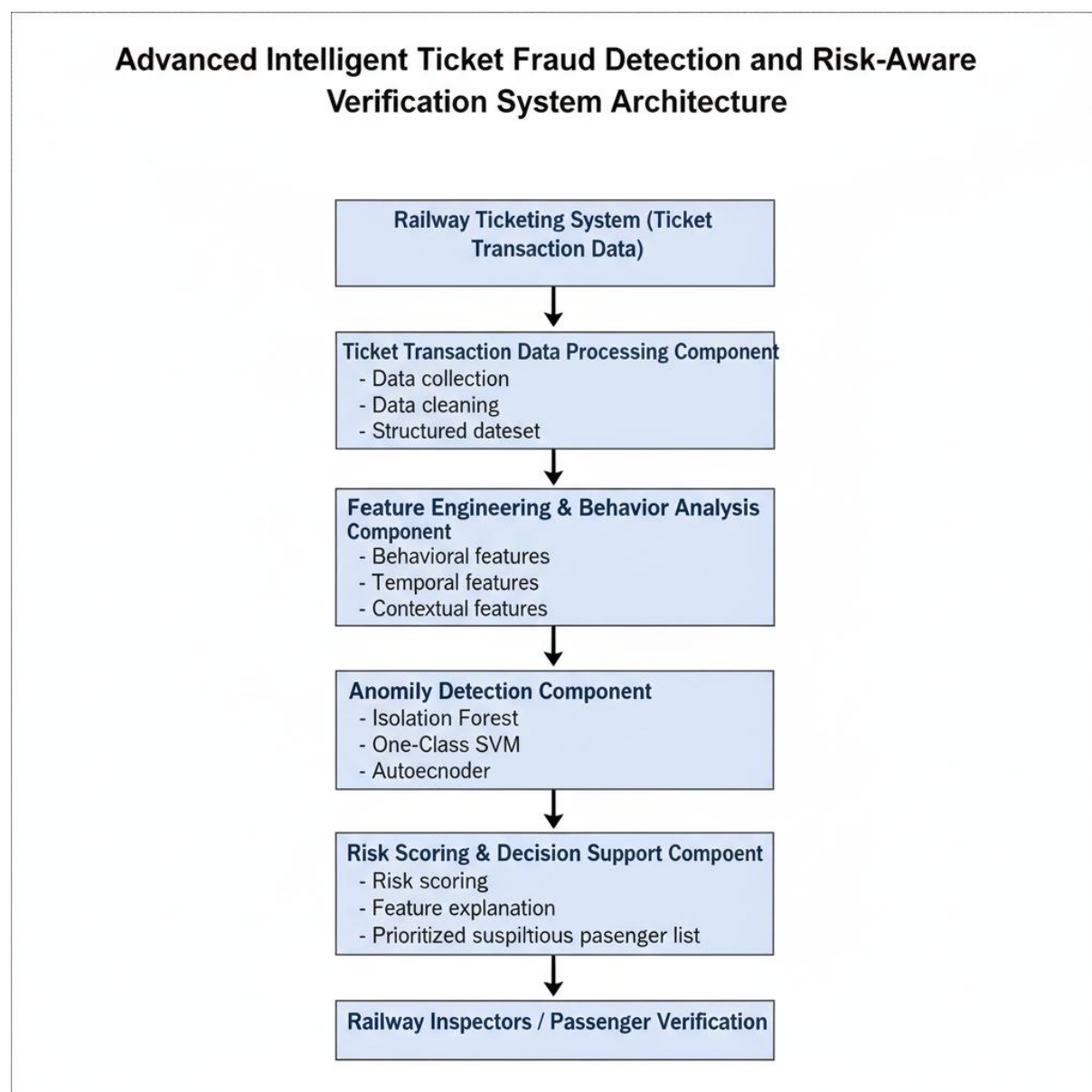


Figure 2: Component overview diagram

5. USER REQUIREMENTS

5.1 Stakeholder Identification

The proposed system involves several stakeholders who interact directly or indirectly with the railway ticket fraud detection framework.

Railway Authorities / Railway Management

The railway authorities are the key stakeholders that manage the railway services effectively by ensuring efficient ticketing services. The authorities need a tool that helps identify fraudulent activities in railway ticketing services. This proposed system helps the railway authorities by providing insights into railway ticketing services and improving the efficiency of ticket inspection services.

Railway Ticket Inspectors

The ticket inspectors are the operational users of the proposed system. The inspectors verify the railway tickets during a train's journey. The proposed framework helps the inspectors by providing information on suspicious passenger transactions. Instead of randomly inspecting the railway tickets, the inspectors can focus on the high-risk passengers that the proposed system has identified. This improves the efficiency of railway inspection services.

Railway Passengers

The railway passengers are the indirect users of the proposed system. The genuine railway passengers benefit from the fair railway inspection services provided by the proposed system. The proposed system helps reduce fraudulent activities in the railway environment. It ensures that genuine railway passengers are treated fairly.

System Administrators and Data Analysts

System administrators and analysts are responsible for the maintenance of the fraud detection system. Additionally, they are required to perform analysis on the ticket transaction data sets. This includes monitoring the efficiency of the anomaly detection models.

5.2 Functional Requirements

The functional requirements describe what the system must do from the perspective of its users, particularly railway inspectors and system administrators:

- 1. Passenger Transaction Monitoring**

The system shall allow inspectors to view real-time and historical ticket transactions to monitor passenger behavior patterns.

- 2. Fraud Detection Alerts**

The system shall automatically flag suspicious ticket transactions based on anomaly

detection models (Isolation Forest, One-Class SVM, Autoencoder) and notify inspectors for verification.

3. **Risk Scoring Dashboard**

The system shall present a risk score for each passenger transaction, along with explanatory information about the features contributing to the anomaly.

4. **Passenger Verification Assistance**

The system shall assist inspectors during ticket checks by providing a prioritized list of high-risk passengers based on risk scoring.

5. **FR5 – Fraud Taxonomy and Dataset Management**

The system shall allow administrators to update and manage the railway fraud taxonomy and maintain synthetic/anonymized ticket transaction datasets for analysis.

6. **FR6 – Reporting and Analytics**

The system shall generate analytical reports summarizing detected fraudulent activities, risk distribution, and inspection effectiveness for decision-making.

5.2 Non-Functional Requirements

The non-functional requirements describe **how the system should perform** to support the functional requirements above:

1. **Performance**

The system shall process and analyze ticket transaction data in real-time, generating risk scores and alerts within 2 seconds per transaction to support inspector decision-making (supports FR2 and FR3).

2. **Security**

The system shall ensure all passenger data is anonymized and encrypted to protect privacy and prevent unauthorized access (supports FR1, FR4, and FR5).

3. **Usability**

The dashboard and alerts shall be user-friendly, allowing inspectors to interpret risk scores and anomaly explanations easily (supports FR2, FR3, FR4).

4. **Scalability**

The system shall handle transaction data from thousands of passengers simultaneously, ensuring system performance does not degrade as usage increases (supports FR1 and FR2).

5. Reliability and Availability

The system shall maintain 99% uptime and recover automatically from failures to ensure continuous monitoring and alerting (supports FR1, FR2, FR3).

6. Maintainability

The system shall allow administrators to update fraud detection models, feature sets, and taxonomy definitions without requiring full system redeployment (supports FR5).

6. COMMERCIALIZATION



1. Customer Segments

Primary Customers: Railway authorities and transportation organizations responsible for passenger ticketing operations.

Customer Persona: Decision-makers in operations, fraud prevention, or IT departments who aim to minimize revenue loss due to ticketing fraud.

Needs: Reliable fraud detection, actionable insights for inspection planning, reduction of manual ticket verification workload.

2. Value Proposition

Provides intelligent, data-driven fraud detection for ticket transactions.

Transforms inspection from random checks to targeted risk-based verification using machine learning anomaly detection.

Generates risk scores for suspicious transactions, offering explainable insights for inspectors.

Reduces revenue loss and improves operational efficiency for railway authorities.

3. Channels

Direct sales to railway authorities through business development teams.

Integration with existing railway ticketing systems via software deployment.

Online demonstrations, webinars, and industry conferences for transportation technology.

4. Customer Relationships

Dedicated support and maintenance: Ensures system reliability and continuous improvement.

Consultative approach: Help organizations understand patterns of fraudulent activity.

Training sessions: For staff to use the dashboard and interpret risk scores.

5. Revenue Streams

Software Licensing: Railway authorities purchase the system as licensed software integrated into their ticketing platforms.

Software-as-a-Service (SaaS): Subscription-based model providing continuous fraud analytics and monitoring.

Custom Analytics Services: Optional advanced services for large transportation authority's needing high-volume transaction analysis.

6. Key Resources

Software components: Fraud detection engine, risk scoring system, anomaly detection algorithms.

Infrastructure: Servers or cloud services for storing and processing ticket transaction data.

Development and maintenance teams: Machine learning engineers, data scientists, and software developers.

7. Key Activities

System Development: Building anomaly detection models and integrating them into railway systems.

Maintenance & Updates: Ensuring system performance and reliability.

Customer Support & Training: Enabling smooth adoption and usage.

8. Key Partnerships

Cloud service providers: For scalable data storage and processing.

Railway IT system vendors: For seamless integration into existing ticketing platforms.

Research institutions: To stay updated on fraud detection and anomaly detection algorithms.

9. Cost Structure

Development Costs: Software engineering, machine learning model development, integration.

Infrastructure Costs: Servers, cloud platforms, database storage.

Maintenance & Support Costs: Ongoing support, updates, and system monitoring.

Marketing & Sales Costs: Promoting the solution to railway authorities and transportation networks.

10. Competitive Advantage

AI-based anomaly detection works without requiring historical fraud labels.

Explainable risk scoring provides clarity for inspectors.

Optimized trade-off between detection efficiency and inspector workload.

Customizable scalable solution for small to large transportation authorities.

11. Intellectual Property

Core IP includes the fraud detection system, risk scoring engine, and anomaly detection workflow.

Protectable via copyrights and software licenses to secure commercialization.

Ensures that railway authorities can safely integrate the software while maintaining ownership of algorithms and processes.

7. BUDGET AND JUSTIFICATION

Table 1: Budget and justification

Category	Item / Description	Estimated Cost (LKR / USD)	Justification
Hardware Costs	Development Workstation (CPU i7, 16GB RAM, 512GB SSD)	LKR 250,000	Required for processing ticket transaction datasets, feature engineering, and training ML models efficiently.
	Additional Storage (External HDD / SSD 1TB)	LKR 30,000	To store datasets, experimental results, and model checkpoints securely.
Software Licenses	Python IDE (PyCharm Professional / Anaconda)	LKR 20,000	Supports development, debugging, and machine learning model implementation.
	ML Libraries / Tools (scikit-learn, TensorFlow, PyTorch)	Free / Open Source	Necessary for implementing Isolation Forest, One-Class SVM, and Autoencoder models.
Cloud Services	Cloud Storage & Compute (Google Cloud / AWS / Azure)	LKR 50,000	Enables scalable processing of large datasets and simulates real-world railway ticketing environment.
Human Effort Cost	Researcher Development & Testing Effort (200 hours @ LKR 1,500/hr)	LKR 300,000	Covers dataset preparation, feature engineering, model implementation, testing, and documentation.
Total Estimated Cost		LKR 650,000	Total investment required to develop, test, and deploy the intelligent ticket fraud detection system.

Hardware Costs

The hardware requirements for the research project mainly include a development workstation that can process the ticket transaction data and perform machine learning model training. A computer system is a general hardware component that can perform data preprocessing and anomaly detection model training for the system development process.

The hardware costs can be justified since machine learning experimentation and data set analysis require reliable hardware for efficient system development.

Software Licenses

The software tools used for the project are the programming tools used for the implementation of the anomaly detection models, as well as the machine learning tools used for the analysis of the data. The tools are used for the implementation of the Isolation Forest, One-Class Support Vector Machine, and Autoencoder models.

Software resources are essential for the research process, as they are used for the development, experimentation, and evaluation of the models.

Cloud Services

Cloud computing environments can be used for the storage of the ticket transaction datasets, as well as for the experimentation of the data processing. The cloud computing environment allows the

system to process large amounts of data, as well as simulate realistic environments for the railway ticketing systems.

Cloud computing is essential for the research process, as it ensures the effectiveness of the fraud detection framework for the transportation industry.

Human Effort Cost

Human effort is the time and effort taken by the researcher to design, implement, test, and evaluate the developed fraud detection system. This includes the preparation of the data set, implementing the model, testing the model experimentally, and documenting the system developed.

Human effort is an important factor since the development of machine learning models and analytical systems demands a lot of research.

8. WORK BREAKDOWN STRUCTURE (WBS)

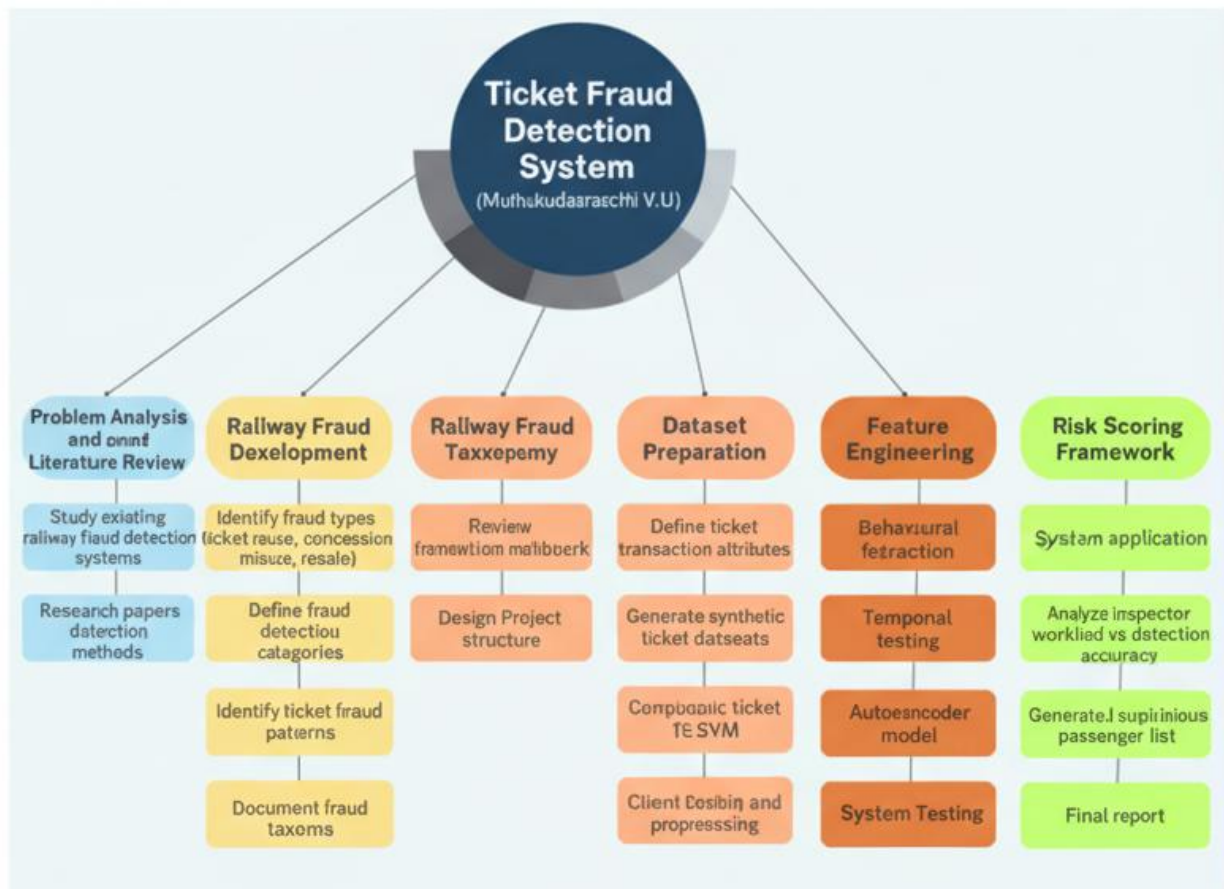


Figure 3: Work breakdown structure

9. GANTT CHART

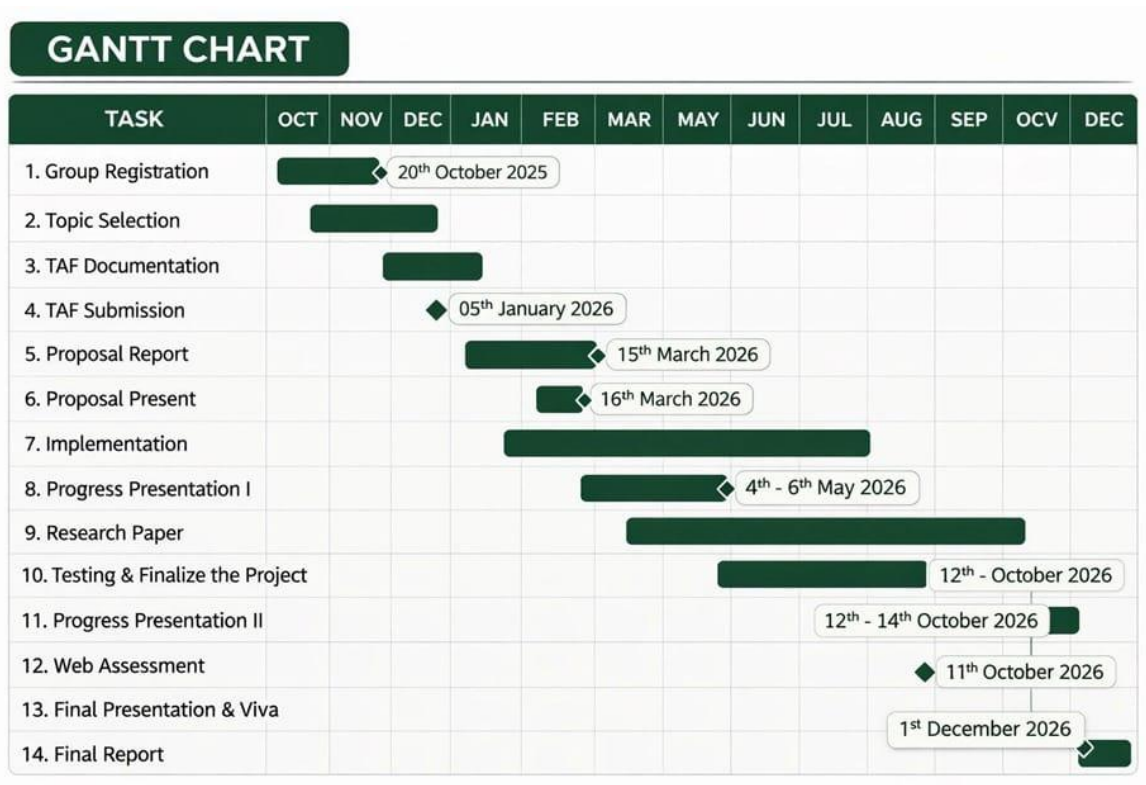


Figure 4: Gantt chart

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11.APPENDICES

11.1 Detailed diagrams

Advanced Intelligent Ticket Fraud Detection
and Risk-Aware Passenger Verification

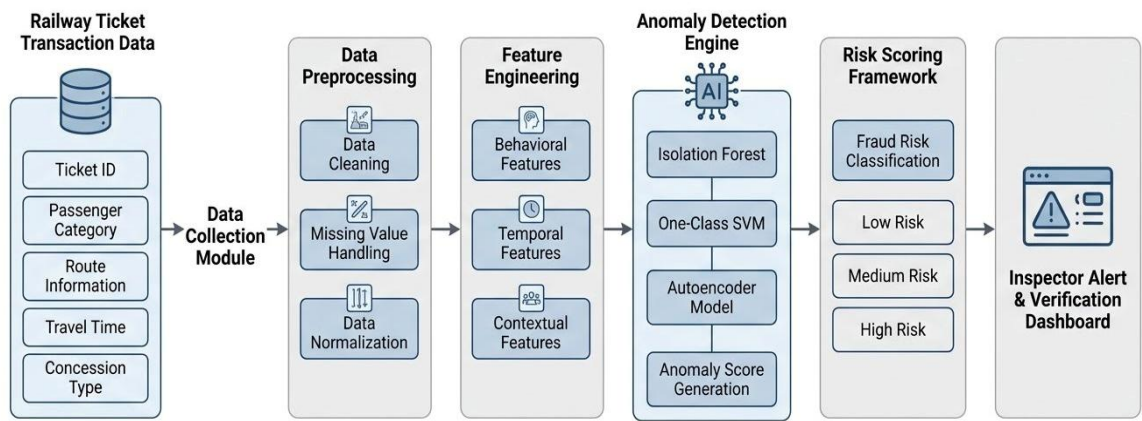


Figure 5: Data processing pipeline diagram

11.2 Preliminary UI mockups

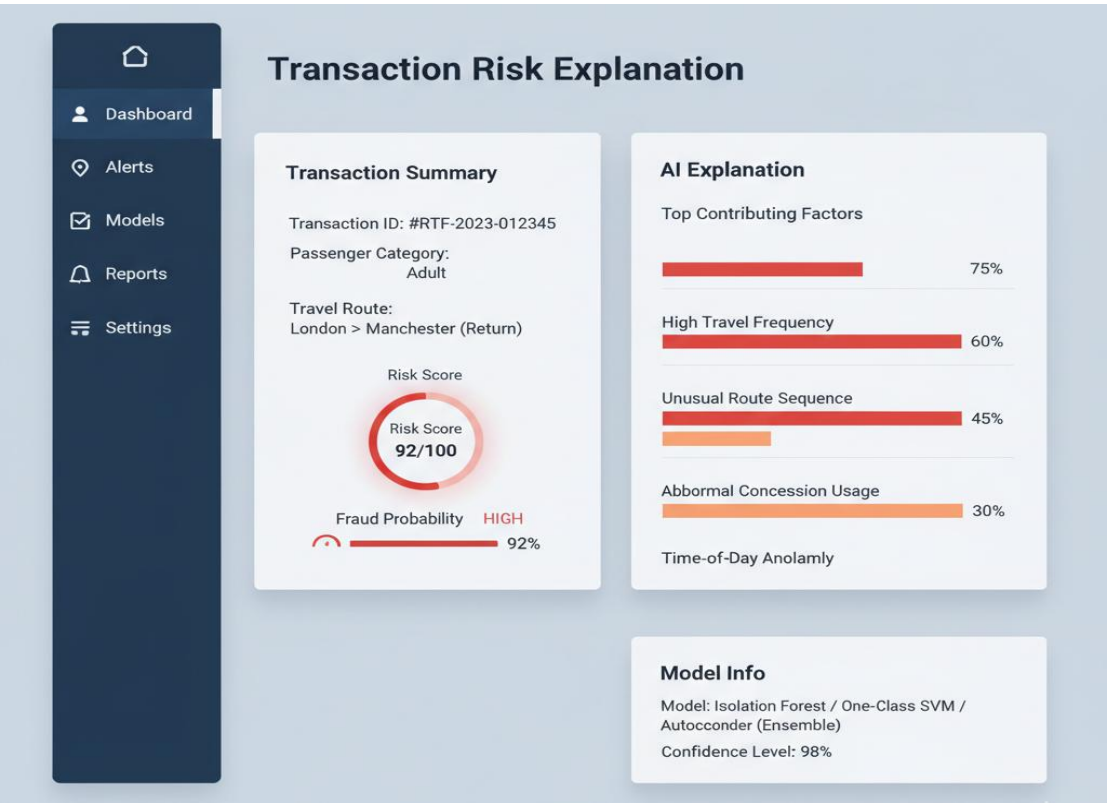


Figure 6: User Interface 1

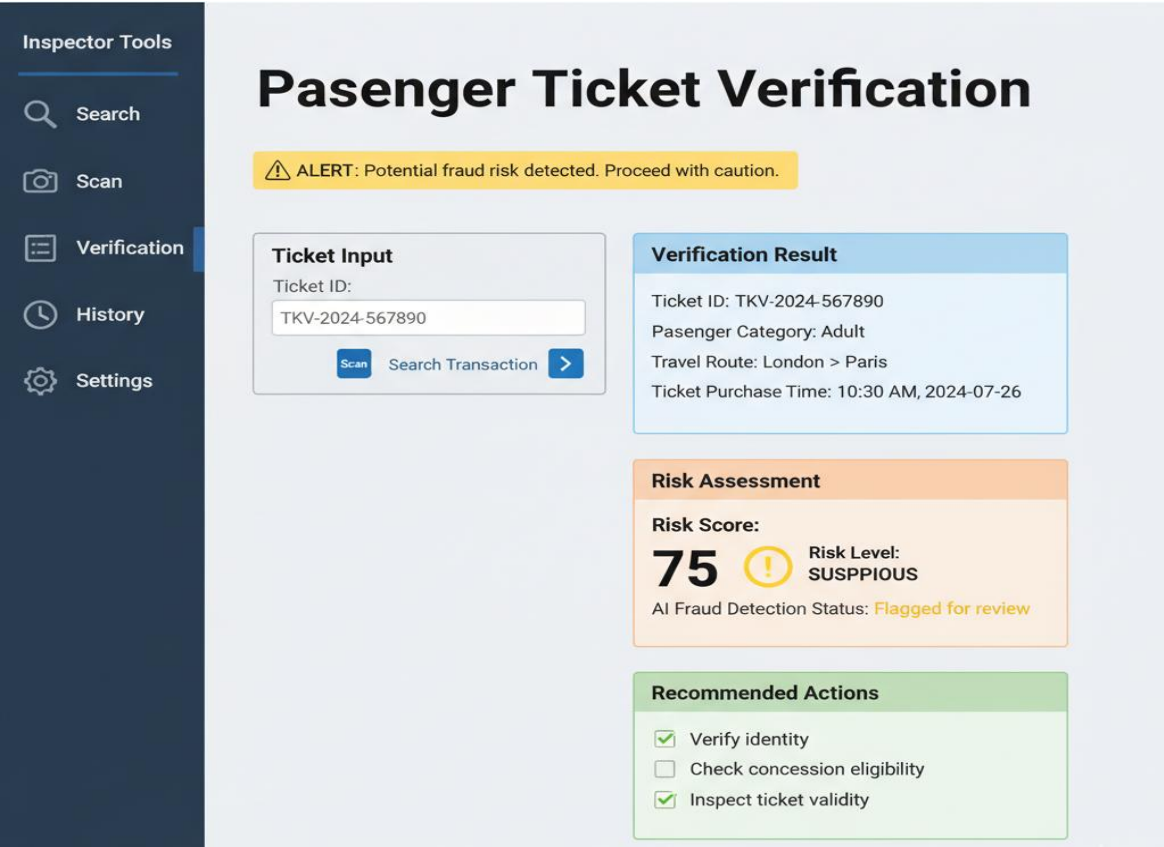


Figure 7: User Interface 2

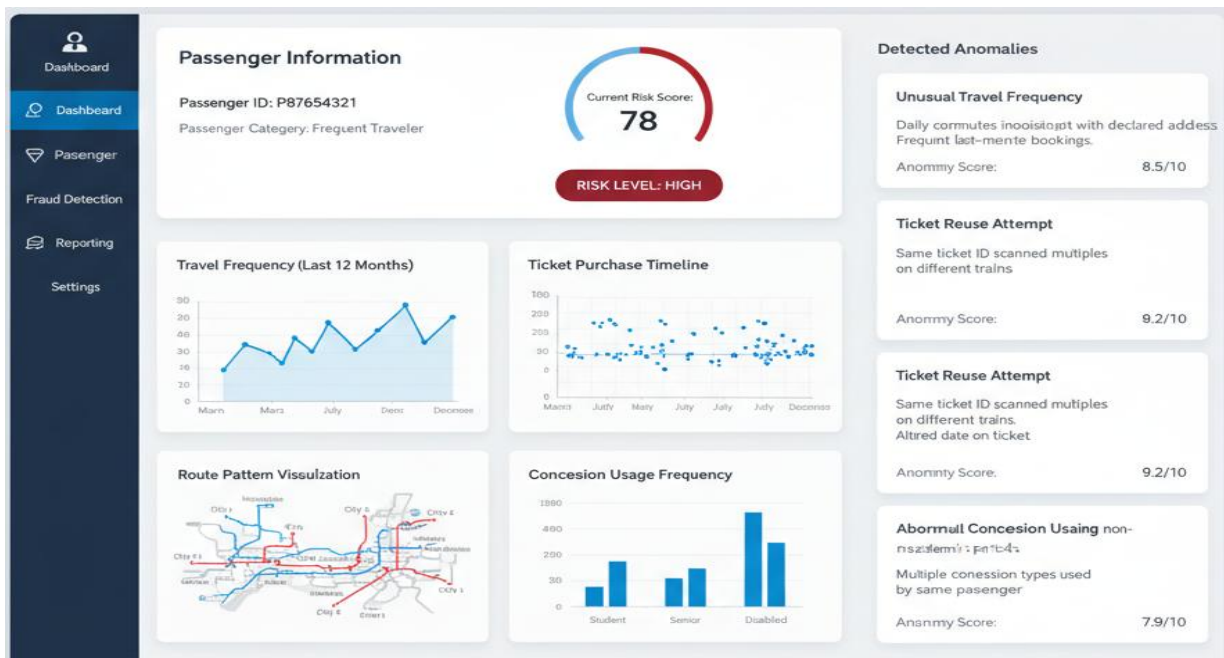


Figure 8: User Interface 3

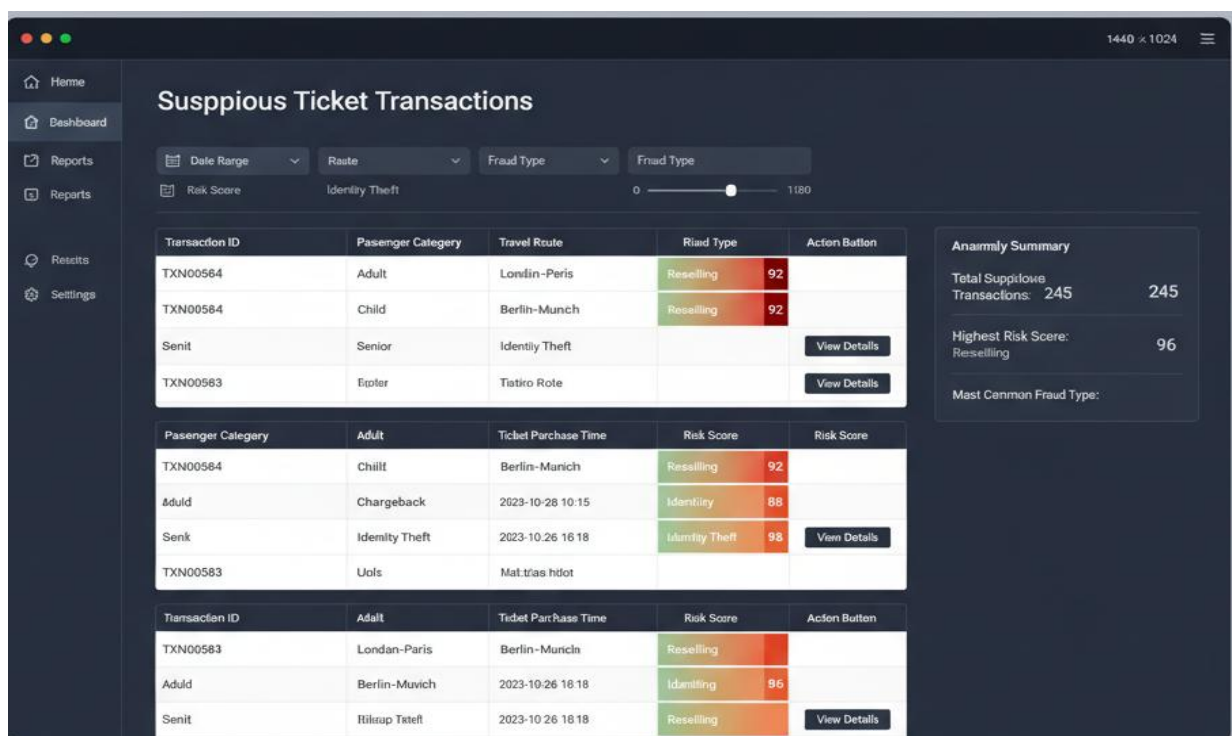


Figure 9: User Interface 4

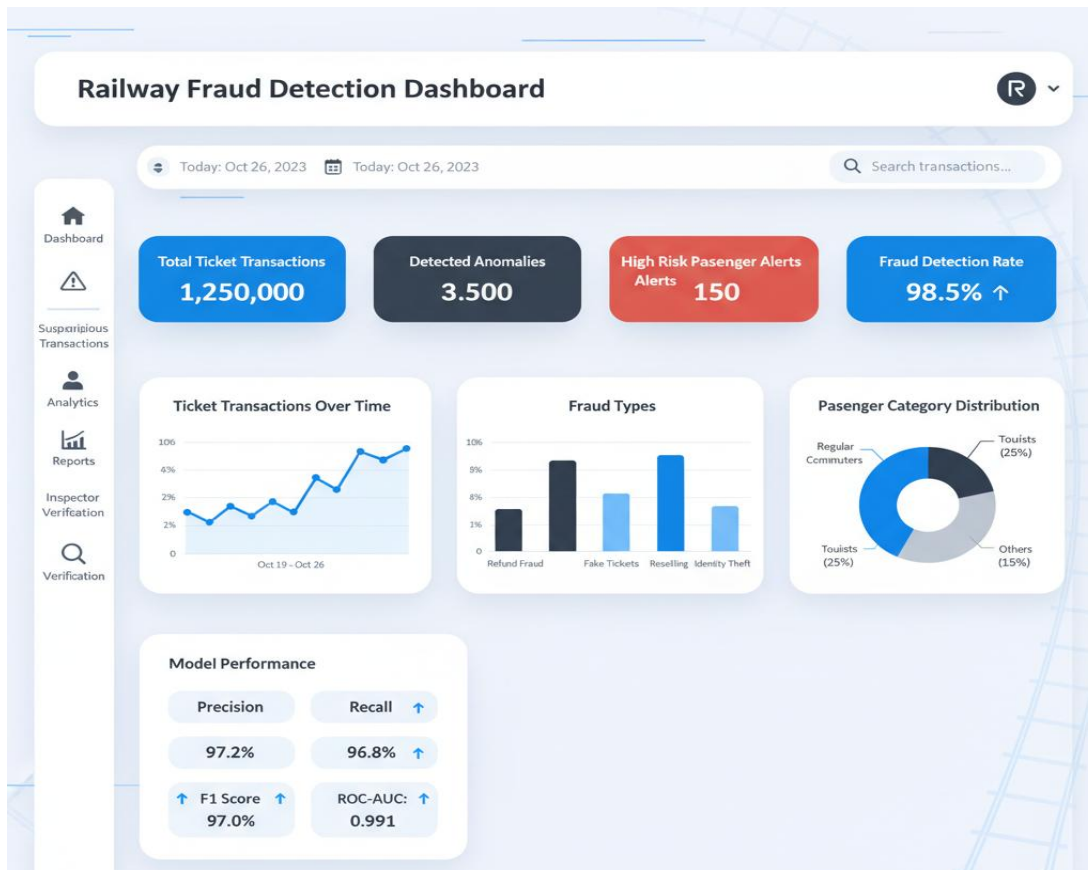


Figure 10: User Interface 5

Passenger Ticket Verification

Ticket Input

Ticket ID:

Verification Result

Passenger Category	Adult - General
Travel Route	London to Edinburgh
Ticket Purchase Time	2024-05-10 14:30

Risk Assessment

Risk Score: 8.5 **Risk Level:** High Fraud Risk

AI Fraud Detection Status: Flagged

Recommended Actions

- ☒ Verify passenger identity with ID
- ☒ Check concession eligibility details
- ☒ Inspect ticket validity and dates

Figure 11: User Interface 6

11.3 Risk analysis

Table 2: Risk analysis

Risk	Impact	Mitigation
Limited availability of real railway ticket fraud datasets	High	Generate anonymized or synthetic ticket transaction datasets that simulate realistic passenger behavior patterns including ticket reuse, concession misuse, and suspicious resale activities.
Imbalanced dataset where fraudulent transactions are rare	High	Apply anomaly detection techniques such as Isolation Forest, One-Class SVM, and Autoencoder models that can detect anomalies without requiring large labeled fraud datasets.
Model accuracy below expected detection performance	High	Perform feature engineering on behavioral, temporal, and contextual ticket data and tune model parameters through iterative experimentation and evaluation.
False positives causing unnecessary passenger inspections	Medium	Implement a risk scoring threshold mechanism and evaluate performance using precision, recall, F1-score, and ROC-AUC to balance fraud detection and inspection workload.
Poor quality or incomplete ticket transaction data	Medium	Apply data preprocessing techniques including data cleaning, missing value handling, and normalization before performing anomaly detection analysis.
Difficulty interpreting model outputs by inspectors	Medium	Develop an explanation framework that highlights anomaly severity and key contributing features influencing suspicious ticket transactions.
Data privacy concerns related to passenger ticket records	Medium	Ensure ticket datasets are anonymized and remove personally identifiable information before data processing and model training.
Integration challenges with existing railway ticketing systems	Medium	Design the system architecture in a modular manner so that the fraud detection component can be integrated with existing ticket transaction databases.
Inspector resistance to adopting automated fraud detection tools	Low	Provide a simple user interface and clear risk explanations to support inspector decision making during ticket verification processes.
Increasing ticket transaction volumes affecting system performance	Medium	Design the system to handle large datasets efficiently and ensure the anomaly detection models are scalable for processing high volumes of ticket transaction data.

11.3 Similarity Report

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